

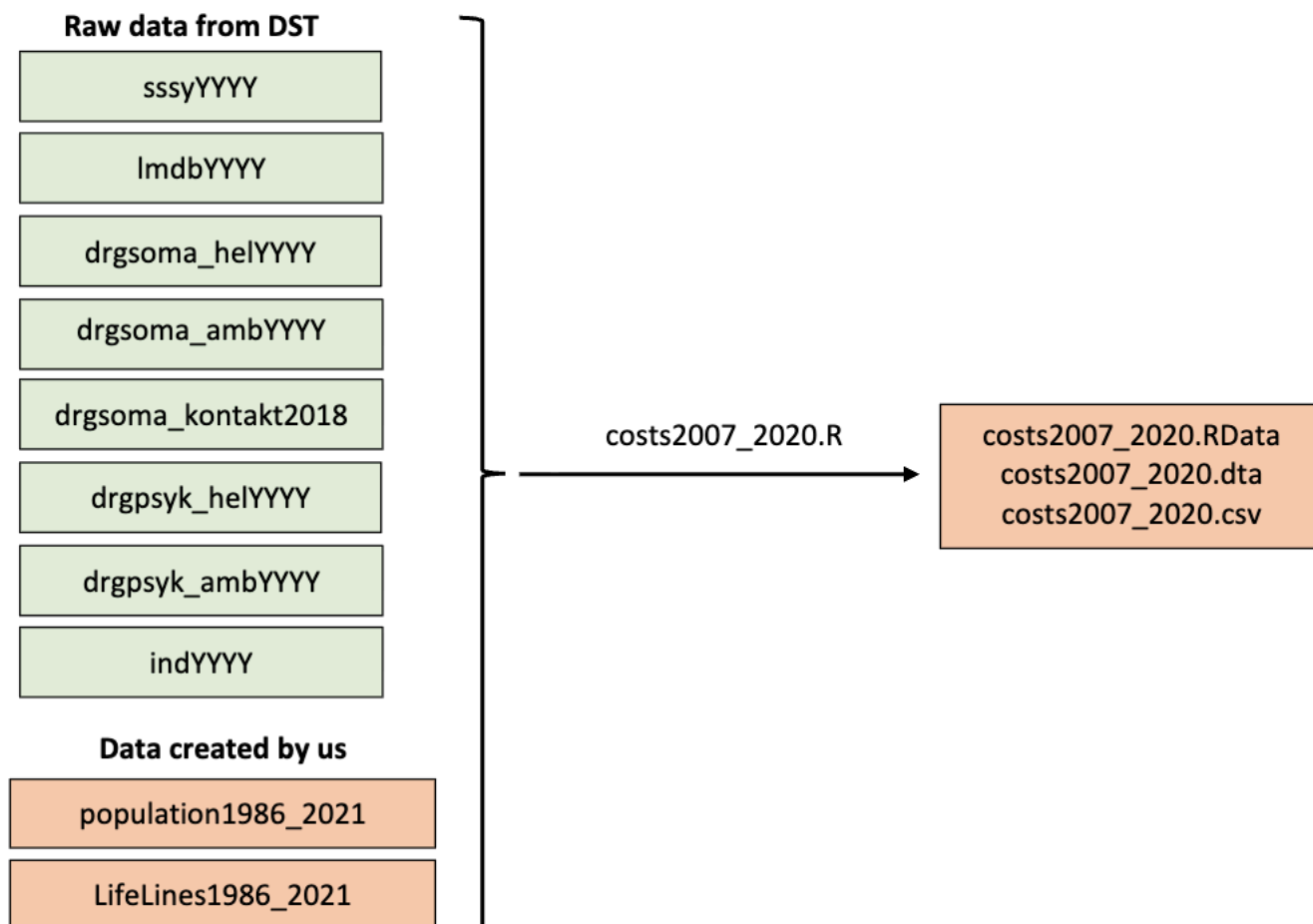
costs2007_2020

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This document is available at Open Science Framework (OSF) (<https://osf.io/dspgg>). If you have any questions or suggestions, or you identify a mistake in our documentation or code, please contact us at lml@clin.au.dk and/or opr@clin.au.dk.

The dataset `costs2007_2020` contains information on healthcare costs, productivity costs, and social transfer payments between January 1, 2007, and December 31, 2020, for all individuals living in Denmark during that period. However, for the years 2019 and 2020, hospital costs are not included. Each person has one observation per year (from 2007 to 2020) with several variables.

Code: `costs2007_2020.R` (R code)
Output: `costs2007_2020.RData` (R file)
`costs2007_2020.dta` (Stata file)
`costs2007_2020.csv` (CSV file)



The dataset is created based on the following registers:

- Register `population1986_2021`, which includes basic information on all individuals living in Denmark between January 1, 1986, and December 31, 2021. We have created this dataset based on the registers from [Statistics Denmark](https://www.ssi.dk) (DST) on

population ([‘Befolkning’; BEF](#)) and deaths in Denmark ([‘Døde i Danmark’; DOD](#)). Specific details and R code are available on OSF (<https://osf.io/dspgq>).

- Register `LifeLines1986_2021`, which includes the life lines of all individuals living in Denmark between January 1, 1986, and December 31, 2021. We have created this dataset with one observation for each period in which someone has lived in Denmark. Each period starts with either birth, immigration to Denmark, or January 1, 1986; and ends with either death, emigration from Denmark, or December 31, 2021. Specific details and R code are available on OSF (<https://osf.io/zhfyp/>).
- The Danish National Health Service Register,¹ which contains information on health services delivered in primary health care by health professionals (e.g., doctors, physiotherapists, and chiropractors) supported by the public/national health insurance since 1990. Until 1996, services provided to children aged less than 16 years were reported using the personal identification number of the mother or father. The register is available through the [Danish Health Data Authority \(Sundhedsdatastyrelsen; SDS\)](#) using [Sygeskriftingsregisteret \(SSR\)](#) or through [Statistics Denmark \(DST\)](#) health service database ([‘Sygeskriftingsstatistik’](#)) with the datasets `sysi` from 1990 to 2005 and the datasets `sssy` from 2006 to 2020.
- The Danish National Prescription Registry,^{2,3} which contains detailed information on prescriptions redeemed at community pharmacies in Denmark since 1995. There are three notable instances when drugs are not filled at community pharmacies and thus are not captured by the registry: (i) drugs used during hospital admissions; (ii) drugs used by certain institutionalized individuals (typically due to psychiatric illnesses); and (iii) drugs supplied directly by hospitals or treatment centers (e.g., chemotherapeutic agents, immunosuppressant drugs, and methadone for those with substance use disorders). The register is available through the [Danish Health Data Authority \(Sundhedsdatastyrelsen; SDS\)](#) through [Lægemiddelstatistikregisteret](#), or through [Statistics Denmark](#) drug database ([‘Lægemiddeldatabasen’; LMDb](#)).
- The DRG National Patient Register ([‘DRG-grupperet LPR’; DRG](#)), which contains information on costs of all admissions to inpatient facilities and visits to outpatient and emergency departments, except hospital services in psychiatric units, based on the Diagnosis Related Group (DRG) and Danish Ambulatory Grouping System (DAGS) tariffs. The DRG and DAGS tariffs are updated each year by the National Board of Health as average national operating expenses for hospital services provided to patients within the same diagnostic group and are used for reimbursement of public and private hospitals.⁴ The register is available through [Statistics Denmark \(DST\)](#) with the datasets `drgsoma_amb` and `drgsoma_hel` from 2005 to 2017 and `drgsoma_kontakt` for 2018.
- The DRG Psychiatric Patient Register, which contains information on costs of all admissions to inpatient facilities and visits to outpatient and emergency departments in psychiatric unit based on the DRG and DAGS tariffs. The DRG and DAGS tariffs are updated each year by the National Board of Health as average national operating expenses for hospital services provided to patients within the same diagnostic group and are used for reimbursement of public and private hospitals.⁴ For psychiatric hospital services, the DRG and DAGS tariffs are averaged according to costs per inpatient bed day and per outpatient visit but do not distinguish between specific diagnoses.⁴ The register is available through [Statistics](#)

[Denmark \(DST\)](#) with the datasets `drgpsyk_hel` and `drgpsyk_amb` from 2005 to 2018.

- Registers on income ([‘Indkomst’; IND](#)), which provides information on all individual social security transfers and taxable income including salaries and financial assets such as stocks, the estimated rental value of own home, etc.

Dataset `costs2007_2020` contains yearly measures of healthcare costs, primary income wage, and social transfers for all individuals. The final dataset contains 79,242,398 observations from 6,973,552 unique identifiers with the following variables:

- **pnr**: Unique personal identifier. All individuals born or immigrating to Denmark receive a unique personal registration number (CPR number) containing 10 digits. Based on different digits’ positions, the CPR number contains information on the date of birth and sex. The [PNR number](#) is a de-identified version of the CPR number made by DST that does not include personal information but allows linkage between different registers based on this unique identifier.
- **year**: Year in which the variables were measured (2007, 2008, ..., 2019, 2020).
- **sex**: Sex based on DST’s variable [koen](#) (1: male; 2: female), indicating the legal sex of the individual based on information from the CPR number.
- **birth_d**: Date of birth based on DST’s variable [foed_dag](#).
- **gp**: Costs of primary healthcare services provided by general practitioners based on DST’s variable [bruhon](#) and [ydtyp](#) (provider code 01 to 10).
- **spec**: Costs of primary healthcare services provided by medical specialists based on DST’s variable [bruhon](#) and [ydtyp](#) (provider code 11 to 35).
- **prsc**: State subsidy costs for prescriptions redeemed at outpatient pharmacies based on DST’s variables [eksp](#) and [ptp](#).
- **som_drg**: Costs of secondary healthcare in somatic hospital departments using the DRG tariffs based on DST’s variable [v_totpris_genop](#) for the years 2007-2017 and [totalpris_drg](#) for year 2018.
- **som_dag**: Costs of secondary healthcare in somatic hospital departments using the DAGS tariffs based on DST’s variable [v_pris_genop](#) for the years 2007-2017 and [totalpris_drg](#) for year 2018.
- **psyc_drg**: Costs of secondary healthcare in psychiatric hospital departments using the DRG tariffs based on DST’s variable [v_totpris](#).
- **psyc_dag**: Costs of secondary healthcare in psychiatric hospital departments using the DAGS tariffs based on DST’s variable [v_pris](#).
- **wage**: Primary taxable wage income (“A-indkomst”) based on DST’s variable [aindk94](#).
- **transfers**: Social transfer payments based on DST’s variable [off_overforsel_13](#). This variable includes unemployment benefits, subsidized employment, sickness allowance, disability pension, old-age pension, early retirement benefit, maternity/paternity pay, integration allowances for refugees, state education grants, housing subsidy, heating aid, green checks, and child benefits.

1. Andersen JS, Olivarius NDF, Krasnik A. The Danish National Health Service Register. *Scand J Public Health*. 2011;39(7 Suppl):34-37. doi:10.1177/1403494810394718
2. Kildemoes HW, Sørensen HT, Hallas J. The Danish National Prescription Registry. *Scand J Public Health*. 2011;39(7 Suppl):38-41. doi:10.1177/1403494810394717

3. Pottegård A, Schmidt SAJ, Wallach-Kildemoes H, Sørensen HT, Hallas J, Schmidt M. Data Resource Profile: The Danish National Prescription Registry. *International Journal of Epidemiology*. 2017;46(3):798-798f. doi:10.1093/ije/dyw213
4. Ankjær-Jensen A, Rosling P, Bilde L. Variable prospective financing in the Danish hospital sector and the development of a Danish case-mix system. *Health Care Manage Sci*. 2006;9(3):259-268. doi:10.1007/s10729-006-9093-1

